

Sai Tarrun Pitta

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Education

California State University, Fullerton

Master of Science in Computer Science

Fullerton, CA

May 2026

GITAM University

Bachelor of Technology in Computer Science

India

June 2023

Experience

Pacific Life

Software Engineer Intern

Newport Beach, CA

June 2025 – March 2026

- Built a serverless **loan-triage service** for commercial and residential underwriting that processed **15,000+ daily records** using Python, AWS Lambda, and LangChain, reducing manual review across underwriting workflows.
- Reduced **loan-policy research time by 96%**, from **15 minutes** of manual lookup to **30 seconds**, by building a citation-backed policy-retrieval service with LangChain and Pinecone.
- Reduced **manual address-processing effort by 75%** by developing a Python and **ESRI REST API** geospatial enrichment pipeline that standardized addresses and generated coordinates for regional analysis.

California State University, Fullerton

Software Developer

Fullerton, CA

September 2024 – May 2025

- Built a **FastAPI-based university knowledge-search platform** indexing **15,000+ documents** across enrollment guides, academic calendars, student handbooks, and IT support resources, enabling semantic, citation-backed search.
- Reduced **median response latency for repeated queries by 70%** by implementing Redis-based semantic caching that reused responses for semantically similar requests.
- Routed up to **70% of low-complexity queries** to quantized open-source LLMs by developing an intent-aware model-routing service that reserved commercial LLM APIs for higher-complexity requests.

Uber

Software Engineer

Hyderabad, India

August 2023 – July 2024

- Reduced checkout timeout rates by **8%** during peak traffic by implementing **Redis-backed distributed session** caching for Uber Eats cart and checkout workflows.
- Reduced **Uber Eats order-history page load time by 22%** by optimizing nested SQL queries and eliminating redundant joins across transactional order tables in **Amazon RDS**.
- Reduced regional **checkout-configuration retrieval latency by 12%** by optimizing Docstore data access and refactoring AWS Lambda request-routing workflows.
- Delivered reusable, accessible React and TypeScript components with **Uber's Base Web design system**, reducing rendering defects by **14%** across global Uber Eats cart and checkout experiences.

Cognizant

Software Engineer

Hyderabad, India

January 2023 – June 2023

- Engineered a **customer-engagement scoring service** for a B2B SaaS platform using Python and scikit-learn, achieving **87% recall** and contributing to a **12% reduction in customer churn**.
- Secured high-traffic e-commerce payment APIs by implementing OpenID Connect, JWT authentication, role-based access control, and Spring Security.
- Engineered SQL-based customer-analytics workflows and **2 normalized reporting tables** in collaboration with 3 business stakeholders and 2 engineers, enabling churn segmentation, campaign tracking, and retention analysis.

Projects

Semantic Code Intelligence Platform

- Built a local-first Python RAG platform indexing **50,000+ lines of code**, combining dense-vector search, BM25, and Cross-Encoder reranking to deliver **sub-second retrieval** with file-path and line-level citations.

Open-SWE Open-Source Contributor

- Delivered **5 merged pull requests** to LangChain's Open-SWE framework by implementing Docker sandbox integration and extending compatibility across multiple large language model providers.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML, CSS

Backend and APIs: Spring Boot, FastAPI, Node.js, Express.js, REST APIs, Microservices, Distributed Systems

Cloud and DevOps: AWS, AWS Lambda, Amazon RDS, Docker, Terraform, Infrastructure as Code, GitHub Actions, CI/CD, Linux

Databases and Retrieval: PostgreSQL, MySQL, MongoDB, Redis, Pinecone, pgvector, BM25, Dense-Vector Search

AI and Frontend: LangChain, Retrieval-Augmented Generation, scikit-learn, Cross-Encoder Reranking, React.js, Base Web

Security and Engineering Practices: OpenID Connect, SAML, JWT, Spring Security, RBAC, System Design, Data Structures and Algorithms, Unit Testing, Performance Testing, Git